

Code for QSS Chapter 1: Introduction

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First Printing

Section 1.1: Overview of the Book

Section 1.2: How to Use this Book

```
install.packages("swirl") # install the package
library(swirl) # load the package
install_course_github("kosukeimai", "qss-swirl") # install the course
library(swirl)
swirl()
```

Section 1.3: Introduction to R

Section 1.3.1: Arithmetic Operations

```
5 + 3
```

```
## [1] 8
```

```
5 - 3
```

```
## [1] 2
```

```
5 / 3
```

```
## [1] 1.666667
```

```
5 ^ 3
```

```
## [1] 125
```

```
5 * (10 - 3)
```

```
## [1] 35
```

```
sqrt(4)
```

```
## [1] 2
```

Section 1.3.2: Objects

```
result <- 5 + 3
result
```

```
## [1] 8
```

```
print(result)
```

```
## [1] 8
result <- 5 - 3
result

## [1] 2
kosuke <- "instructor"
kosuke

## [1] "instructor"
kosuke <- "instructor and author"
kosuke

## [1] "instructor and author"
Result <- "5"
Result

## [1] "5"
result

## [1] 2
class(result)

## [1] "numeric"
Result

## [1] "5"
class(Result)

## [1] "character"
class(sqrt)

## [1] "function"
```

Section 1.3.3: Vectors

```
world.pop <- c(2525779, 3026003, 3691173, 4449049, 5320817, 6127700, 6916183)
world.pop

## [1] 2525779 3026003 3691173 4449049 5320817 6127700 6916183
pop.first <- c(2525779, 3026003, 3691173)
pop.second <- c(4449049, 5320817, 6127700, 6916183)
pop.all <- c(pop.first, pop.second)
pop.all

## [1] 2525779 3026003 3691173 4449049 5320817 6127700 6916183
world.pop[2]

## [1] 3026003
world.pop[c(2, 4)]

## [1] 3026003 4449049
```

```

world.pop[c(4, 2)]

## [1] 4449049 3026003
world.pop[-3]

## [1] 2525779 3026003 4449049 5320817 6127700 6916183
pop.million <- world.pop / 1000
pop.million

## [1] 2525.779 3026.003 3691.173 4449.049 5320.817 6127.700 6916.183
pop.rate <- world.pop / world.pop[1]
pop.rate

## [1] 1.000000 1.198047 1.461400 1.761456 2.106604 2.426063 2.738238
pop.increase <- world.pop[-1] - world.pop[-7]
percent.increase <- (pop.increase / world.pop[-7]) * 100
percent.increase

## [1] 19.80474 21.98180 20.53212 19.59448 15.16464 12.86752
percent.increase[c(1, 2)] <- c(20, 22)
percent.increase

## [1] 20.00000 22.00000 20.53212 19.59448 15.16464 12.86752

```

Section 1.3.4: Functions

```

length(world.pop)

## [1] 7
min(world.pop)

## [1] 2525779
max(world.pop)

## [1] 6916183
range(world.pop)

## [1] 2525779 6916183
mean(world.pop)

## [1] 4579529
sum(world.pop) / length(world.pop)

## [1] 4579529
year <- seq(from = 1950, to = 2010, by = 10)
year

## [1] 1950 1960 1970 1980 1990 2000 2010

```

```

seq(to = 2010, by = 10, from = 1950)

## [1] 1950 1960 1970 1980 1990 2000 2010
seq(from = 2010, to = 1950, by = -10)

## [1] 2010 2000 1990 1980 1970 1960 1950
2008:2012

## [1] 2008 2009 2010 2011 2012
2012:2008

## [1] 2012 2011 2010 2009 2008
names(world.pop)

## NULL
names(world.pop) <- year
names(world.pop)

## [1] "1950" "1960" "1970" "1980" "1990" "2000" "2010"
world.pop

##      1950      1960      1970      1980      1990      2000      2010
## 2525779 3026003 3691173 4449049 5320817 6127700 6916183

## myfunction <- function(input1, input2, ..., inputN) {
##
##     DEFINE `output' USING INPUTS
##
##     return(output)
## }

my.summary <- function(x){ # function takes one input
  s.out <- sum(x)
  l.out <- length(x)
  m.out <- s.out / l.out
  out <- c(s.out, l.out, m.out) # define the output
  names(out) <- c("sum", "length", "mean") # add labels
  return(out) # end function by calling output
}
z <- 1:10
my.summary(z)

##      sum length  mean
##   55.0   10.0   5.5

my.summary(world.pop)

##      sum  length  mean
## 32056704       7 4579529

```

Section 1.3.5: Data Files

```
## setwd("qss/INTRO")
## getwd()

UNpop <- read.csv("UNpop.csv")
class(UNpop)

## [1] "data.frame"
load("UNpop.RData")

names(UNpop)

## [1] "year"      "world.pop"
nrow(UNpop)

## [1] 7
ncol(UNpop)

## [1] 2
dim(UNpop)

## [1] 7 2
summary(UNpop)

##      year      world.pop
##  Min.   :1950   Min.     :2525779
##  1st Qu.:1965   1st Qu.  :3358588
##  Median :1980   Median  :4449049
##  Mean   :1980   Mean    :4579529
##  3rd Qu.:1995   3rd Qu.  :5724258
##  Max.   :2010   Max.     :6916183
UNpop$world.pop

## [1] 2525779 3026003 3691173 4449049 5320817 6127700 6916183
UNpop[, "world.pop"] # extract the column called "world.pop"

## [1] 2525779 3026003 3691173 4449049 5320817 6127700 6916183
UNpop[c(1, 2, 3),] # extract the first three rows (and all columns)

##   year world.pop
## 1 1950   2525779
## 2 1960   3026003
## 3 1970   3691173
UNpop[1:3, "year"] # extract the first three rows of the "year" column

## [1] 1950 1960 1970
## take elements 1, 3, 5, ... of the "world.pop" variable
UNpop$world.pop[seq(from = 1, to = nrow(UNpop), by = 2)]

## [1] 2525779 3691173 5320817 6916183
```

```

world.pop <- c(UNpop$world.pop, NA)
world.pop

## [1] 2525779 3026003 3691173 4449049 5320817 6127700 6916183      NA
mean(world.pop)

## [1] NA
mean(world.pop, na.rm = TRUE)

## [1] 4579529

```

Section 1.3.6: Saving Objects

```

## save.image("qss/INTRO/Chapter1.RData")

## save(UNpop, file = "Chapter1.RData")
## save(world.pop, year, file = "qss/INTRO/Chapter1.RData")

## write.csv(UNpop, file = "UNpop.csv")

## load("Chapter1.RData")

```

Section 1.3.7: Packages

```

install.packages("foreign") # install package
library("foreign") # load package

read.dta("UNpop.dta")
read.spss("UNpop.sav")

write.dta(UNpop, file = "UNpop.dta")

```

Section 1.3.8: Programming and Learning Tips

```

source("UNpop.R")

##
## File: UNpop.R
## Author: Kosuke Imai
## The code loads the UN population data and saves it as a STATA file
##

library(foreign)
UNpop <- read.csv("UNpop.csv")
UNpop$world.pop <- UNpop$world.pop / 1000 # population in millions
write.dta(UNpop, file = "UNpop.dta")

library(lintr)
lint("UNpop.R")

```

